

		$v = 200 \text{ m/s}$
6	A	Force by wall on water = $\langle F_{net} \rangle = \text{mass per unit time} \times \Delta v$ $= \frac{m}{\Delta t} \Delta v = \frac{\rho(\text{Vol})}{\Delta t} \Delta v \quad \text{since } m = \rho(\text{Vol})$ $= \rho \pi r^2 v (0 - v)$ $= -\rho \pi r^2 v^2$ $= -1000 \pi (2.0 \times 10^{-2})^2 (0.50)^2$ $= -0.314 \text{ N}$ By Newton's 3rd Law, Force by water on wall = Force by wall on water = 0.314 N
7	D	On each cuboid :